

Gene Expression Signatures in Hepatobiliary Carcinomas

Differential Diagnosis and the Role of Hepatic Stellate Cell Activation

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Abstract

Hepatobiliary carcinomas (HBCs), including hepatocellular carcinoma and cholangiocarcinoma, are aggressive malignancies with limited diagnostic and therapeutic options. The tumor microenvironment, particularly hepatic stellate cell (HSC) activation, plays a critical role in disease progression by promoting fibrosis, immune modulation, and extracellular matrix remodeling. In this study, transcriptomic data from the GSE148355 dataset were analyzed to uncover gene expression signatures associated with HBC and HSC activity. Data preprocessing, normalization, and principal component analysis revealed distinct transcriptional clusters. Differential gene expression analysis using DESeq2 identified key genes, including GRIA4, DKK1, and KCNU1, associated with neuroimmune signaling and fibrotic pathways. Enrichment analyses via ILINCS, Enrichr, DAVID, and Reactome confirmed the involvement of Wnt signaling, neurotransmission, and vesicular transport mechanisms. Network analyses using GeneMANIA and PiiNET highlighted functionally interconnected modules and upstream regulatory enzymes. Together, these findings provide a comprehensive molecular profile of HBC, emphasize the role of hepatic stellate cell-mediated remodeling, and propose novel biomarkers and therapeutic targets for future investigation.

Introduction

Hepatobiliary carcinomas (HBCs), comprising hepatocellular carcinoma (HCC) and cholangiocarcinoma (CCA), are among the most challenging and lethal malignancies worldwide, accounting for significant cancer-related morbidity and mortality. Despite considerable advances in surgical techniques, liver transplantation protocols, and systemic therapies, the overall five-year survival rate for HBC remains dismal. A major factor contributing to this poor prognosis is the frequent late-stage diagnosis, when curative options are limited. Early stages of the disease are often clinically silent, and distinguishing benign hepatic lesions from malignant ones based solely on imaging and histopathology remains difficult due to overlapping morphological features. Consequently, there is a critical unmet need for reliable molecular diagnostic markers and mechanistic understanding that can enable early detection, differential diagnosis, and personalized therapeutic interventions.

The pathogenesis of hepatobiliary malignancies is a complex and multifactorial process, involving genetic alterations, chronic inflammation, fibrosis, and remodeling of the extracellular matrix (ECM). Among the non-parenchymal cells implicated in liver fibrosis and carcinogenesis, hepatic stellate cells (HSCs) have gained significant attention. Under physiological conditions, HSCs remain in a quiescent state, storing vitamin A and contributing minimally to ECM production. However, upon activation by inflammatory cues, viral infections, metabolic insults, or oncogenic signaling, HSCs transdifferentiate into myofibroblast-like cells. Activated HSCs secrete large quantities of collagen, laminin, fibronectin, and other ECM components, leading to fibrosis. Furthermore, HSCs influence immune cell infiltration, promote immunosuppressive niches, stimulate angiogenesis, and directly facilitate tumor cell survival, migration, and proliferation through paracrine signaling mechanisms.

Given the central role of HSCs in fibrosis and tumor microenvironment remodeling, understanding the molecular signatures associated with their activation and interaction with tumor cells has immense diagnostic and therapeutic potential. However, the transcriptional landscape governing HSC behavior in the context of hepatobiliary carcinomas remains incompletely understood. Most existing studies focus separately on cirrhosis, hepatitis, or HCC, without systematically integrating multi-layered transcriptomic and network-based analyses to map HSC-driven changes at the systems level.

Recent advancements in high-throughput technologies, particularly gene expression microarrays and RNA sequencing, have revolutionized our ability to profile tumors at the molecular level. Publicly available repositories such as the Gene Expression Omnibus (GEO) now house large datasets that can be mined to uncover novel biomarkers, signaling pathways, and therapeutic targets. Integrating these datasets with sophisticated bioinformatics platforms allows for comprehensive characterization of differential gene expression patterns, pathway dysregulation, and molecular network perturbations in disease states.

In this study, we leveraged transcriptomic data from the GEO dataset GSE148355, focusing on hepatobiliary carcinoma samples. Our objective was to identify gene expression patterns specific to malignant versus non-malignant tissues, particularly genes involved in HSC activation, fibrosis, and tumor microenvironment modulation. We employed an integrative bioinformatics workflow comprising data preprocessing, normalization, principal component analysis (PCA), and differential expression analysis using the DESeq2 statistical framework.

Following DEG identification, we conducted multi-platform functional enrichment analyses using ILINCS, Enrichr, DAVID, and Reactome databases to annotate biological pathways, cellular compartments, and molecular functions enriched among significant genes. Additionally, ToppGene Suite was utilized for a comprehensive multi-ontology enrichment, including disease associations, drug-gene interactions, transcription factor binding motifs, and microRNA targets. We further mapped protein-protein interaction (PPI) networks using GeneMANIA and visualized gene-centric interactomes and post-translational modification networks using PiiNET and its PTMs2Modifiers module.

This holistic approach aims to dissect the underlying biological processes contributing to hepatobiliary carcinoma progression, emphasizing the role of hepatic stellate cell-driven remodeling of the tumor microenvironment. We hypothesize that specific gene expression signatures associated with HSC activation correlate with disease aggressiveness, immune evasion, and extracellular matrix alterations, and that mapping these networks can reveal novel biomarkers for early diagnosis as well as therapeutic intervention points.

Furthermore, by integrating transcriptomic data with systems biology and network analysis tools, this study seeks to bridge the gap between raw genomic data and actionable biological insights. In doing so, we contribute to the growing field of precision oncology, where multi-omic profiling informs risk stratification, treatment selection, and disease monitoring strategies tailored to individual patient molecular profiles.

In summary, this research integrates high-dimensional transcriptomic datasets with advanced bioinformatics and network biology methodologies to identify and characterize gene expression signatures in hepatobiliary carcinomas, with a special focus on hepatic stellate cell activity. The outcomes of this study are expected to enhance our understanding of HBC pathogenesis, provide molecular candidates for further validation, and open avenues for developing HSC-targeted therapies aimed at intercepting fibrosis-driven carcinogenesis.

Materials and Methods

● 2.1 Data Acquisition & Pre-processing

Gene expression data for hepatobiliary carcinoma samples were retrieved from the Gene Expression Omnibus (GEO) database, specifically from the dataset **GSE148355**. GEO2R, an NCBI-provided web tool, was used to access and analyze the dataset. Sample groups, including control and tumor tissues, were defined based on available metadata. The full expression matrix, containing probe IDs and expression values, was downloaded in text format for downstream processing.

● 2.2 Probe Annotation and Gene Symbol Extraction

The platform annotation file associated with the dataset (GPL file) was downloaded from the GEO database to map probe IDs to corresponding gene symbols. The annotation file included mappings between probe identifiers (ID_REF) and gene symbols (GENE_SYMBOL). Probe IDs from the expression matrix were merged with gene symbol information using Excel, ensuring that each expression value could be accurately assigned to its respective gene.

● 2.3 Data Cleaning and Normalization

Following probe-to-gene mapping, the dataset was subjected to cleaning processes. Redundant probes, non-annotated entries, and missing values were removed. For genes represented by multiple probes, the probe with the highest mean expression value was retained to ensure consistency. After cleaning, the dataset was normalized and transformed into **FPKM (Fragments Per Kilobase Million)** format to allow comparison of gene expression across samples.

● 2.4 Uploading into Bioinformatics Platform

The cleaned, annotated, and normalized expression dataset was uploaded to the **T-BioInfo platform** for advanced analyses. T-BioInfo provides integrative pipelines for dimensionality reduction, differential expression analysis, and network-based interpretations.

● 2.5 Principal Component Analysis (PCA)

● Dimensionality Reduction by PCA :

Principal Component Analysis (PCA) was performed on the normalized dataset using the unsupervised learning module of T-BioInfo (PCA_R_Library block). PCA enabled the reduction of high-dimensional gene expression data into principal components, highlighting the main sources of variance between samples. Both two-dimensional and three-dimensional PCA plots were generated for visual interpretation of sample clustering patterns.

● 2.6 Differential Gene Expression Analysis (D-Seq Analysis)

● Unsupervised Machine Learning

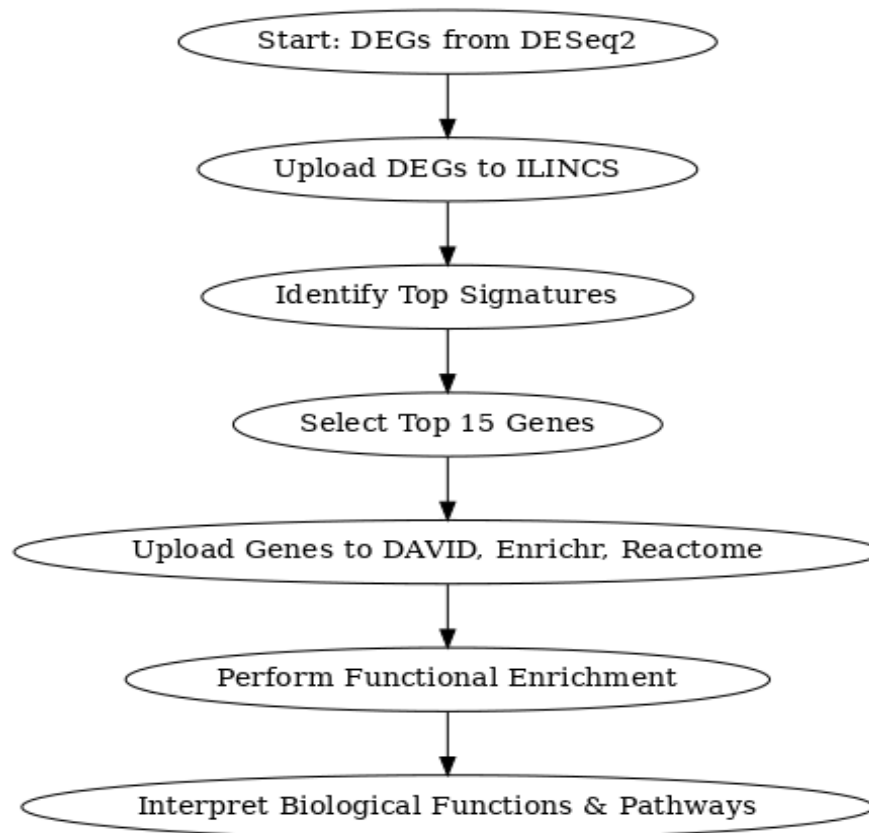
Unsupervised differential gene expression analysis was performed using the DESeq2 pipeline. MA plot, log fold change (LFC) plot, and volcano plot were used to visualize the magnitude and significance of gene regulation.

● Selection of Significant Genes

From the DESeq2 output (dseq.txt), genes were filtered using adjusted p-value < 0.05 and $|\log_2FC| > 1$. The resulting significant genes were further submitted to multiple enrichment and annotation tools in *ILinc*.

● 2.7 Functional Annotation

- The Integrative LINCS (ILINCS) platform is a comprehensive bioinformatics tool designed to explore transcriptional responses to various perturbations, including chemical compounds, genetic modifications, and disease states. ILINCS integrates gene expression profiles from the Library of Integrated Network-Based Cellular Signatures (LINCS) program, enabling researchers to match user-defined gene signatures with known perturbation-induced transcriptional changes. By comparing user-uploaded differential expression data against reference signatures, ILINCS facilitates the identification of pathways, biological processes, and potential therapeutic agents associated with specific gene expression patterns.
- In this study, ILINCS was utilized to analyze the differentially expressed genes (DEGs) derived from the GSE148355 dataset. A filtered list of DEGs, based on statistical significance (adjusted p-value < 0.05, $|\log_2FC| > 1$), was uploaded to ILINCS for signature matching. The platform identified top-ranking gene signatures & perturbations, which were subsequently subjected to pathway and functional enrichment analyses using integrated resources such as DAVID, Enrichr & Reactome.



- **Functional Enrichment and Pathway Analysis**

- Gene signatures were analyzed using:**

1. **ILINCS**: For identifying consistent disease/perturbation signatures.
2. **DAVID**: Functional Annotation and Gene Ontology clustering.
3. **Enrichr**: KEGG, GO Biological Process, and Molecular Function analyses.
4. **Reactome**: Pathway mapping using curated human biological interactions.
5. **ToppGene**: Multi-category enrichment including disease, domain, phenotype, and microRNA.

- **Network and Interaction Analysis**

- GeneMANIA** was used to explore protein-protein interactions, co-expression, and co-localization patterns. Predicted genes were included for functional enrichment of subnetworks.

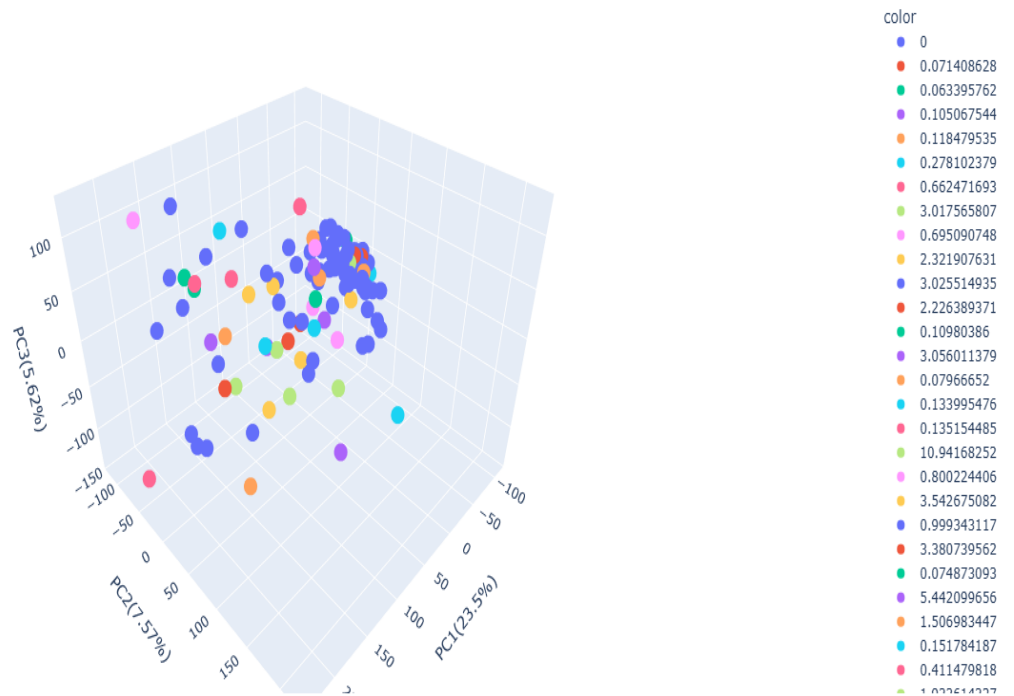
- **Post-Translational Modification & Visualization**

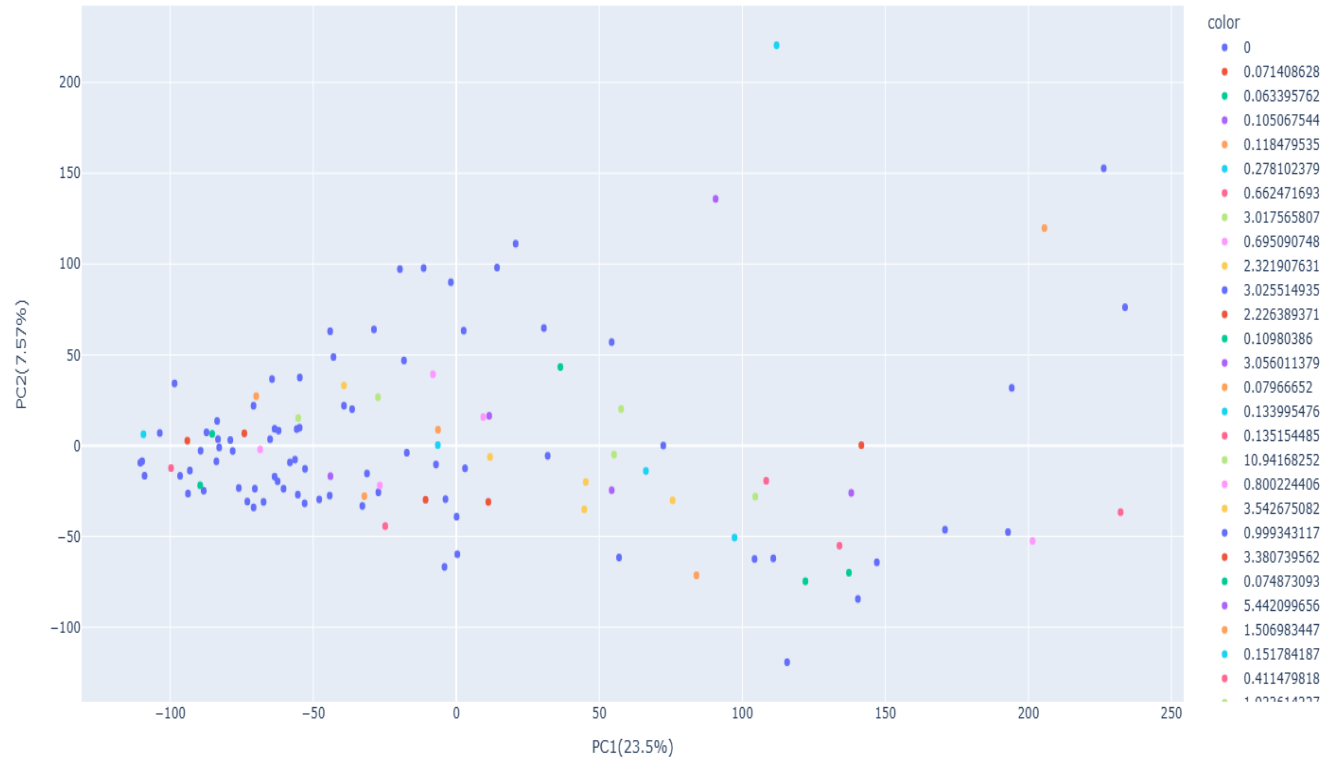
- PiiNET** and its **PTMs2Modifiers** module were employed to investigate upstream modifying enzymes and visualize pathway-centric networks. This provides insight into post-transcriptional regulatory elements governing gene expression patterns.

● 2.8 Results

Principal Component Analysis (PCA)

PCA was conducted using T-BioInfo's unsupervised learning module. Three principal components (PC1, PC2, and PC3) captured significant variance in the expression data. PCA 2D and 3D plots demonstrated clear separation of sample clusters, reflecting distinct transcriptomic profiles across hepatobiliary carcinoma samples.

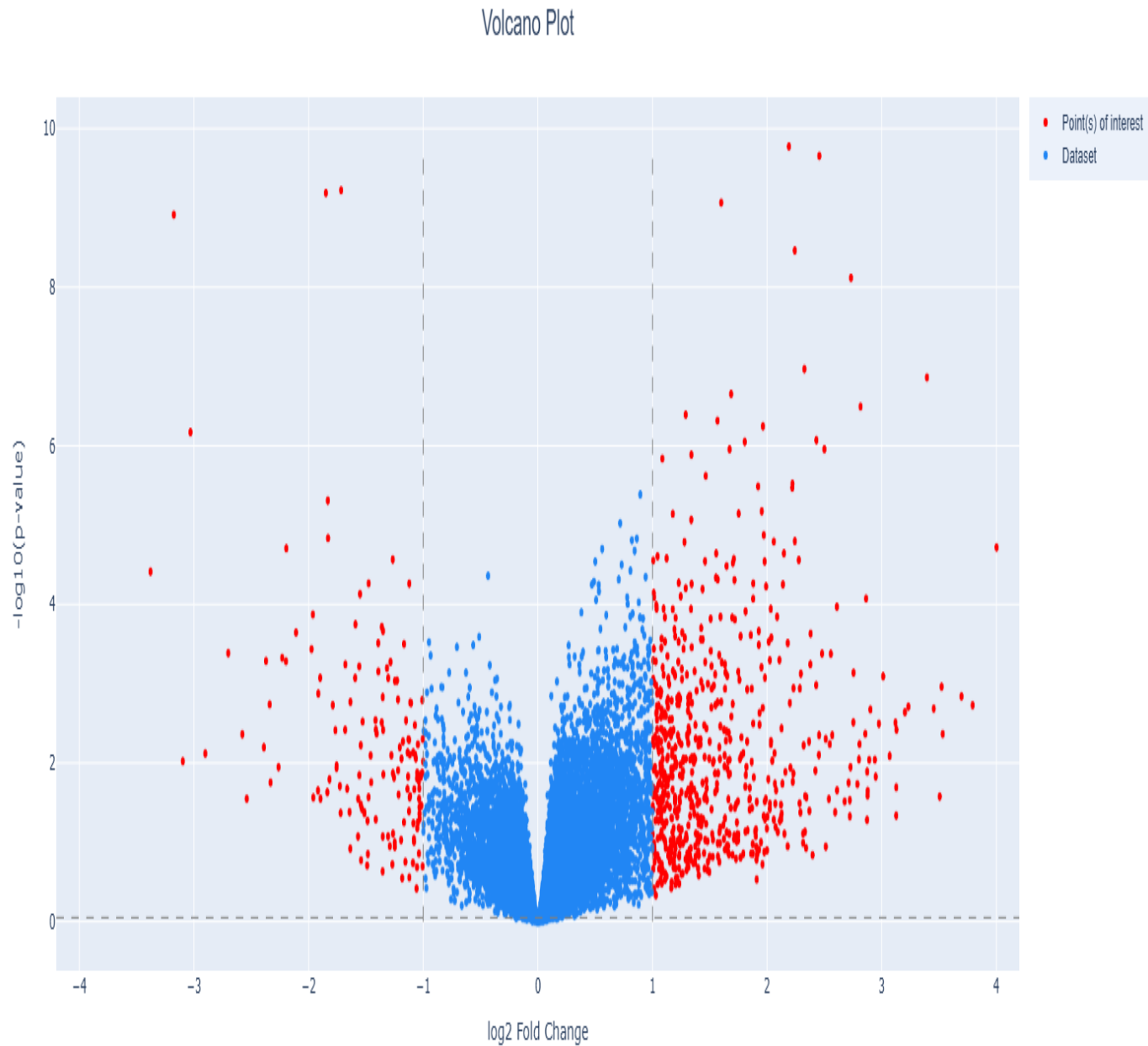


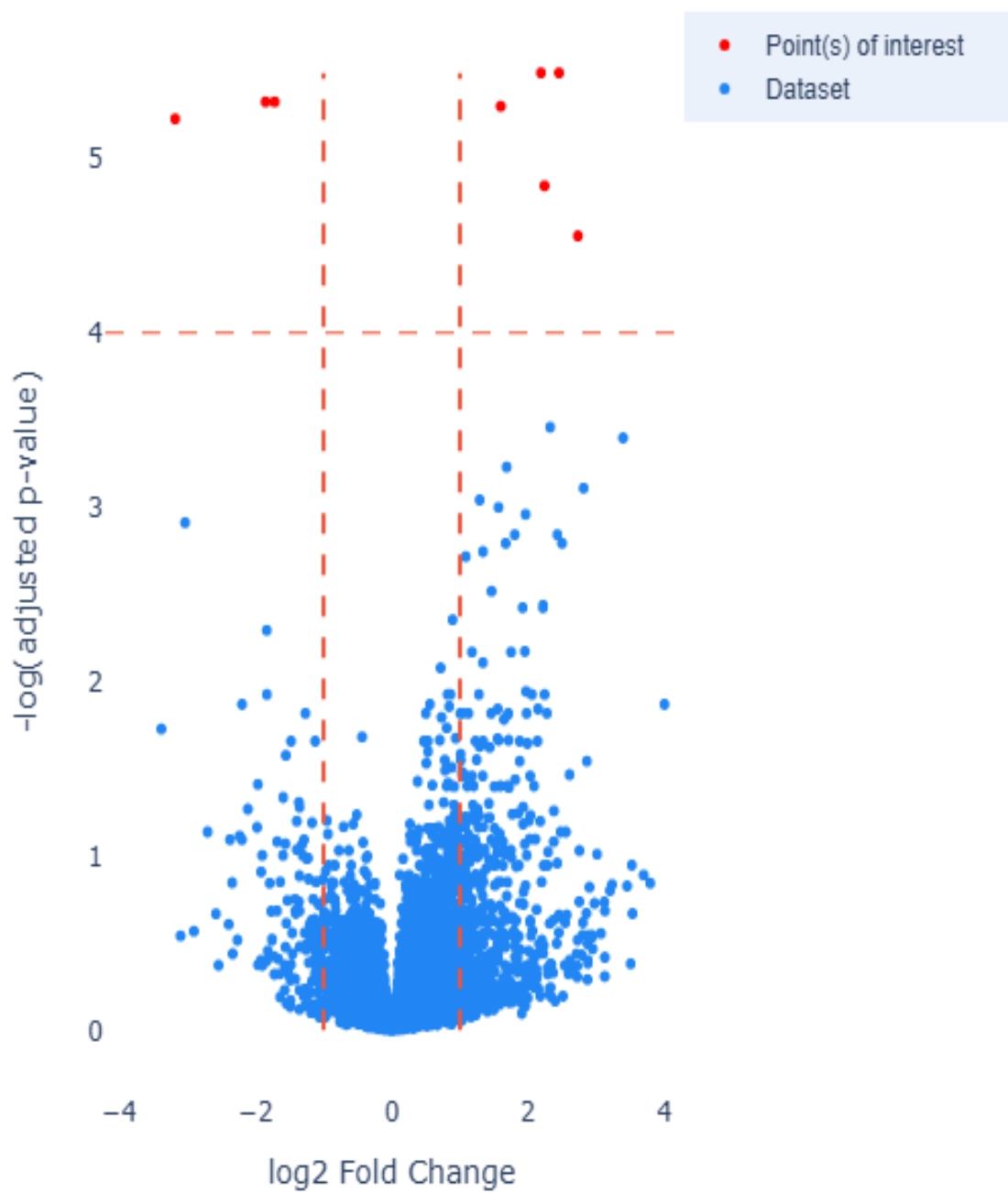


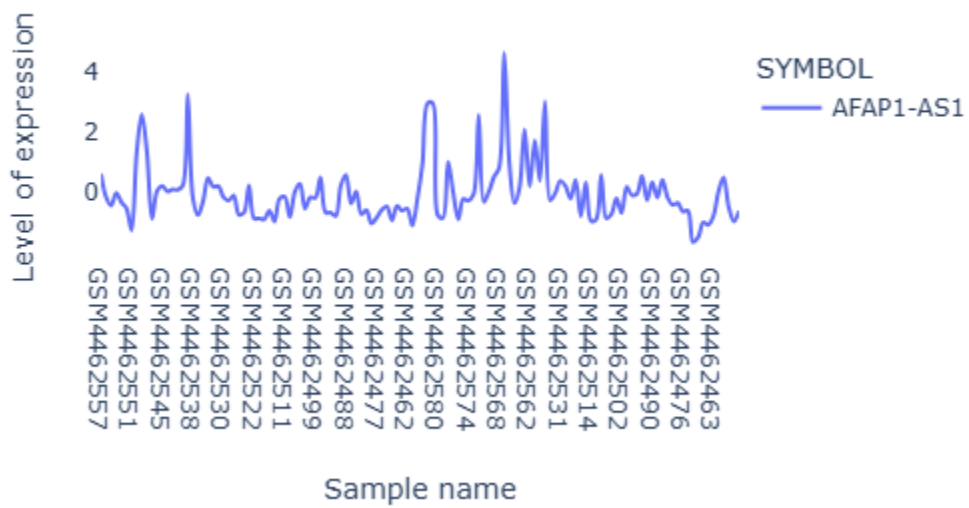
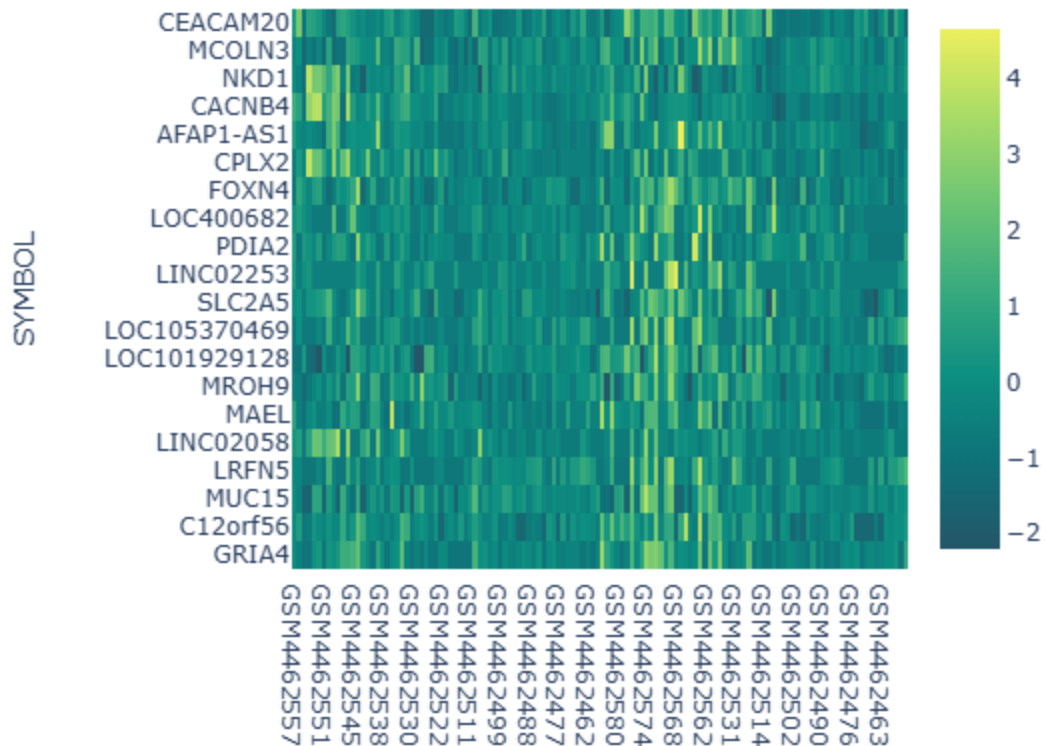
Name	PC1	PC2	PC3
GSM4462456	-58.14	-9.2	36.58
GSM4462457	-55.38	-26.94	37.35
GSM4462458	-76.02	-23.3	21.93
GSM4462459	-28.78	64	16.1
GSM4462460	-54.75	9.9	22.94
GSM4462461	-56.38	-7.68	32.35
GSM4462462	-52.87	-12.76	25.65
GSM4462463	-63.5	-17.05	28.24
GSM4462464	-94.02	2.76	1.89
GSM4462465	-36.41	20.11	25.26
GSM4462466	-88.29	-24.79	15.46
GSM4462467	-60.38	-23.7	37.64
GSM4462468	-108.97	-16.56	-7.09
GSM4462469	-89.52	-21.82	13.34
GSM4462470	-73.04	-30.74	-0.25
GSM4462471	110.82	-62.04	57.11
GSM4462472	-17.29	-3.89	2.87
GSM4462473	-31.21	-15.32	3.65
GSM4462474	-32.83	-33.16	17.41
GSM4462475	11.52	16.45	38.39
GSM4462476	-6.38	8.76	16.71
GSM4462477	30.62	64.75	27.9
GSM4462478	0.11	-39.1	32.68
GSM4462479	66.29	-13.92	-35.96

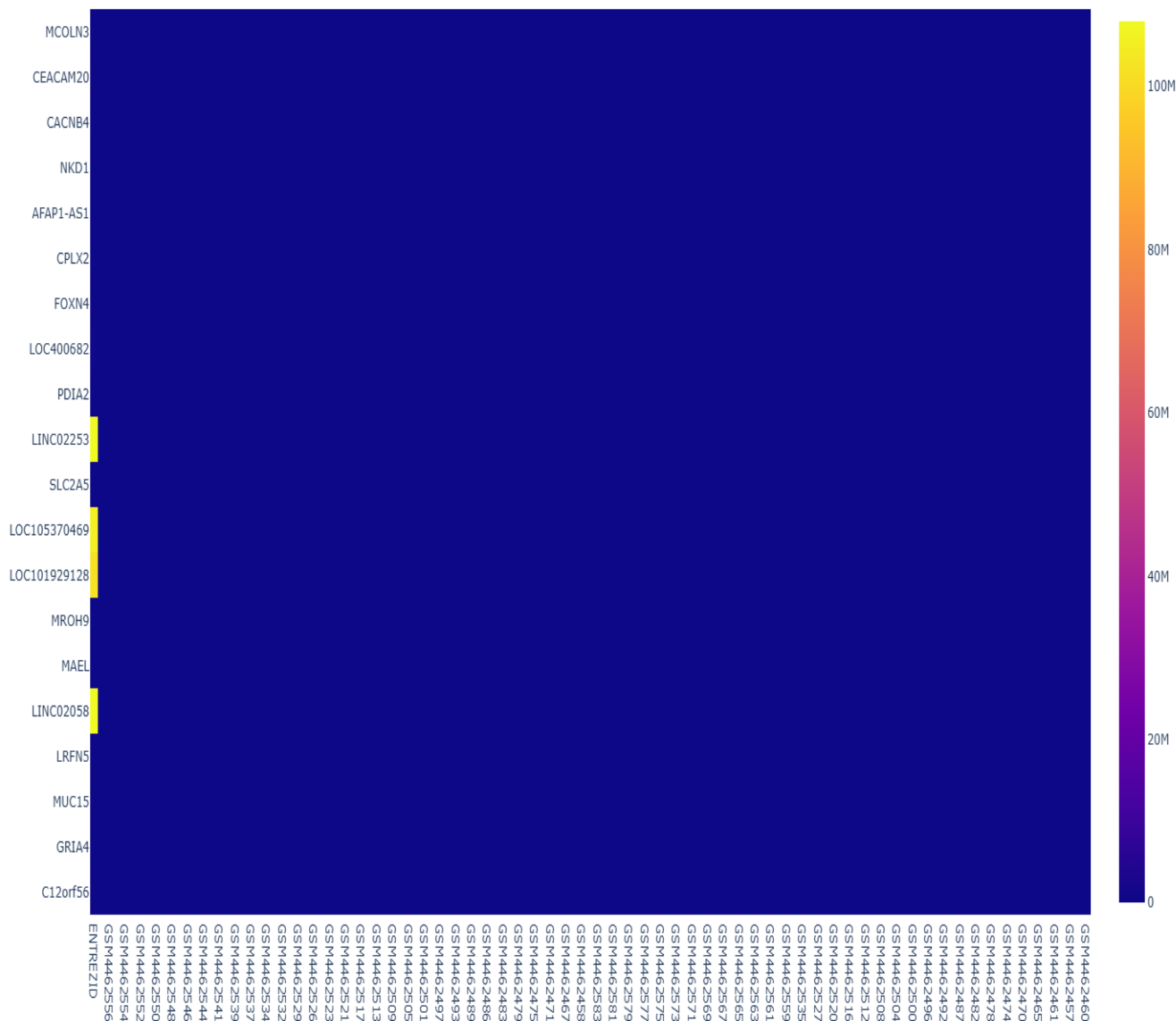
4. Differential Gene Expression (DESeq2)

DESeq2 analysis yielded a list of statistically significant DEGs (cutoff: adjusted $p < 0.05$, $|\log_2\text{FC}| > 1$). MA plot, LFC distribution, and Volcano plots revealed numerous up- and down-regulated genes.









5. Gene Expression Visualization

Heatmaps of top 20 DEGs and single-gene plots (e.g., AFAP1-AS1) were generated. The clustered heatmap revealed consistent expression across distinct samples, while single-gene traces illustrated heterogeneity across tissues.

6. Functional Enrichment & ILINCS Signature Analysis

ILINCS signature analysis identified 15 high-confidence DEGs. These were subjected to GO, KEGG in Enrichr, Reactome, and DAVID enrichment, revealing biological pathways associated with immune signaling, neurotransmission, and fibrosis.

GO Biological Process Enrichment table :

Term	Genes	Adjusted P-value	Combined Score
Positive Regulation of Wnt Signaling Pathway, Planar Cell Polarity Pathway (GO:2000096)	DKK1	0.0459603235094198	1993.609466758919
Negative Regulation of Cardiac Muscle Cell Differentiation (GO:2000726)	DKK1	0.0459603235094198	1993.609466758919
Cell Junction Disassembly (GO:0150146)	DKK1	0.0459603235094198	1993.609466758919
Negative Regulation of Cardiocyte Differentiation (GO:1905208)	DKK1	0.0459603235094198	1542.8675927558304

Synapse Organization (GO:0050808)	LRFN5;DKK1	0.04596032350941 98	113.550095795412 91
Long-Chain Fatty Acid Catabolic Process (GO:0042758)	GLYATL2	0.04596032350941 98	1249.0776854949
Heart Induction (GO:0003129)	DKK1	0.04596032350941 98	1249.0776854949
Protein Retention in ER Lumen (GO:0006621)	PDIA2	0.04596032350941 98	1249.0776854949
Synapse Pruning (GO:0098883)	DKK1	0.04596032350941 98	1043.43429845216 4
Regulation of Heart Morphogenesis (GO:2000826)	DKK1	0.04596032350941 98	1043.43429845216 4
Regulation of Wnt Signaling Pathway, Planar Cell Polarity Pathway (GO:2000095)	DKK1	0.04596032350941 98	1043.43429845216 4
Regulation of Cardiac Muscle Cell Differentiation (GO:2000725)	DKK1	0.04596032350941 98	892.013120036697

Negative Regulation of Smooth Muscle Cell Apoptotic Process (GO:0034392)	GRIA4	0.04596032350941 98	776.212425317664
Positive Regulation of Non- Canonical Wnt Signaling Pathway (GO:2000052)	DKK1	0.04596032350941 98	776.212425317664
Maintenance of Protein Localization in Endoplasmic Reticulum (GO:0035437)	PDIA2	0.04596032350941 98	685.007341093276 5
Negative Regulation of Muscle Cell Apoptotic Process (GO:0010656)	GRIA4	0.04596032350941 98	500.500391445587 5
Medium-Chain Fatty Acid Metabolic Process (GO:0051791)	GLYATL2	0.04596032350941 98	500.500391445587 5
Negative Regulation of Macrophage Activation (GO:0043031)	LRFN5	0.04596032350941 98	457.732815153479 7

Regulation of Cell Fate Specification (GO:0042659)	DKK1	0.0459603235094198	457.7328151534797
Regulation of Smooth Muscle Cell Apoptotic Process (GO:0034391)	GRIA4	0.0459603235094198	421.0918230074216
Regulation of Presynapse Assembly (GO:1905606)	LRFN5	0.0459603235094198	421.0918230074216
Regulation of Presynapse Organization (GO:0099174)	LRFN5	0.0459603235094198	421.0918230074216
Synaptic Membrane Adhesion (GO:0099560)	LRFN5	0.0459603235094198	389.38044082427126
Negative Regulation of Striated Muscle Cell Differentiation (GO:0051154)	DKK1	0.0459603235094198	389.38044082427126
Mesenchyme Development (GO:0060485)	DKK1	0.0467009976298178	361.6911269349327

Signaling Receptor Ligand Precursor Processing (GO:0140448)	MEP1A	0.04738295973054 94	337.323781189166 3
Negative Regulation of Leukocyte Activation (GO:0002695)	LRFN5	0.04913732038432 24	296.474504982362 3
Negative Regulation of SMAD Protein Signal Transduction (GO:0060392)	DKK1	0.04913732038432 24	279.207318677517 66
Endocardial Cushion Development (GO:0003197)	DKK1	0.04913732038432 24	279.207318677517 66
Synaptic Transmission, Glutamatergic (GO:0035249)	GRIA4	0.05424899524745 07	225.039973613912 1
Limb Development (GO:0060173)	DKK1	0.05424899524745 07	225.039973613912 1
Peptide Hormone Processing (GO:0016486)	MEP1A	0.05871928706446 05	186.995957660366 2

Negative Regulation of Ossification (GO:0030279)	DKK1	0.05871928706446 05	186.995957660366 2
Regulation of Macrophage Activation (GO:0043030)	LRFN5	0.0588714341317	179.212381370595 72
Glutamate Receptor Signaling Pathway (GO:0007215)	GRIA4	0.06622341457230 39	147.555100876472 35
Regulation of Receptor Internalization (GO:0002090)	DKK1	0.06622341457230 39	142.378481464209 13
Positive Regulation of Macromolecule Biosynthetic Process (GO:0010557)	DKK1;FER1L6	0.06659698880282 54	30.4947216125598 54
Negative Regulation of Proteolysis (GO:0045861)	TM4SF20	0.06775387645849 38	128.596741196650 7
Regulation of Receptor- Mediated Endocytosis (GO:0048259)	DKK1	0.06926908204290 7	120.640460027010 4

Regulation of Ossification (GO:0030278)	DKK1	0.0707040802347871	113.499815495736
Regulation of Trans-Synaptic Signaling (GO:0099177)	GRIA4	0.0736056998751608	104.0719468696984
Negative Regulation of Protein Binding (GO:0032091)	DKK1	0.0763596485153274	95.91771596133704
Regulation of Proteolysis (GO:0030162)	TM4SF20	0.0789762468439077	88.80359995755282
Positive Regulation of Gene Expression (GO:0010628)	DKK1;FER1L6	0.0815987041494308	22.09006807811813
Negative Regulation of Protein-Containing Complex Assembly (GO:0031333)	DKK1	0.0815987041494308	77.0124855419272
Regulation of SMAD Protein Signal Transduction (GO:0060390)	DKK1	0.0815987041494308	75.30634662640054

Fatty Acid Catabolic Process (GO:0009062)	GLYATL2	0.08159870414943 08	75.3063466264005 4
Regulation of Synapse Assembly (GO:0051963)	LRFN5	0.08510667351934 29	69.0845479012675
Negative Regulation of Binding (GO:0051100)	DKK1	0.08591526113611 64	66.2937955640167 3
Regulation of Neuron Differentiation (GO:0045664)	DKK1	0.09661725495578 85	54.8166489621250 4
Regulation of Protein Binding (GO:0043393)	DKK1	0.09714789913188 07	52.9206203994512 45
Positive Regulation of JNK Cascade (GO:0046330)	DKK1	0.09884110153363 22	50.2762294491887 8
Long-Chain Fatty Acid Metabolic Process (GO:0001676)	GLYATL2	0.09974178375149 62	47.0803282057625 8
Actomyosin Structure Organization (GO:0031032)	MYO18B	0.09974178375149 62	47.0803282057625 8

Regulation of Wnt Signaling Pathway (GO:0030111)	DKK1	0.1230958330583807	32.17211116568681
Regulation of JNK Cascade (GO:0046328)	DKK1	0.1230958330583807	31.004592702550895
Protein Processing (GO:0016485)	MEP1A	0.1230958330583807	31.004592702550895
Modulation of Chemical Synaptic Transmission (GO:0050804)	GRIA4	0.1230958330583807	30.26450036869356
Negative Regulation of Canonical Wnt Signaling Pathway (GO:0090090)	DKK1	0.1230958330583807	30.26450036869356
Negative Regulation of Defense Response (GO:0031348)	LRFN5	0.1230958330583807	28.868690726427467
Negative Regulation of Inflammatory Response (GO:0050728)	LRFN5	0.1230958330583807	28.868690726427467

- Reg of Transmembrane Receptor Prot Serine/Threonine Kinase Sgnlng Pway (GO:0090101)	DKK1	0.12309583305838 07	28.5362170293238 4
Potassium Ion Transport (GO:0006813)	KCNU1	0.12578587042856 03	26.9642434603322 1
Regulation of Neuron Apoptotic Process (GO:0043523)	DKK1	0.12578587042856 03	26.3747864855391 33
Potassium Ion Transmembrane Transport (GO:0071805)	KCNU1	0.12842859929170 82	24.9898833832729 6
Negative Regulation of Protein Metabolic Process (GO:0051248)	TM4SF20	0.13519023076692 54	22.1102420875820 2
Negative Regulation of Wnt Signaling Pathway (GO:0030178)	DKK1	0.13519023076692 54	22.1102420875820 2
Negative Regulation of Response to External Stimulus (GO:0032102)	LRFN5	0.14011288248338 04	20.4773944915012 8

Heart Development (GO:0007507)	DKK1	0.15025016923334 45	17.5909448832875 5
Cell-Cell Adhesion via Plasma-Membrane Adhesion Molecules (GO:0098742)	LRFN5	0.15025016923334 45	17.5909448832875 5
Response to Endoplasmic Reticulum Stress (GO:0034976)	PDIA2	0.16833421599776 51	14.3896651942717 12
Regulation of Canonical Wnt Signaling Pathway (GO:0060828)	DKK1	0.18168038450449 75	12.4449080358892 11
Negative Regulation of Multicellular Organismal Process (GO:0051241)	DKK1	0.19973072514309 88	10.4089773680844 77
Regulation of Inflammatory Response (GO:0050727)	LRFN5	0.20148182752846 43	10.0270958510300 8
Chemical Synaptic Transmission (GO:0007268)	GRIA4	0.20679689802769 42	9.38120981195214 8

Negative Regulation of Signal Transduction (GO:0009968)	DKK1	0.2161585438160547	8.502403022934251
Negative Regulation of Intracellular Signal Transduction (GO:1902532)	DKK1	0.2252718105042765	7.621096522393087
Monoatomic Cation Transmembrane Transport (GO:0098655)	KCNU1	0.2252718105042765	7.345377121468697
Regulation of Gene Expression (GO:0010468)	DKK1;FER1L6	0.2252718105042765	4.077834279838571
Inorganic Cation Transmembrane Transport (GO:0098662)	KCNU1	0.2252718105042765	7.2318443605558675
Positive Regulation of MAPK Cascade (GO:0043410)	DKK1	0.2574891749627598	5.545649843072898
Negative Regulation of Programmed Cell Death (GO:0043069)	DKK1	0.2665883519522118	5.076452619713442

Negative Regulation of Apoptotic Process (GO:0043066)	DKK1	0.3173874809346231	3.5126251480790005
Regulation of Apoptotic Process (GO:0042981)	DKK1	0.4307682954682337	1.718719840207129
Negative Regulation of Transcription by RNA Polymerase II (GO:0000122)	DKK1	0.4385861372941219	1.5943830503164806
Negative Regulation of DNA-templated Transcription (GO:0045892)	DKK1	0.5452937704827604	0.8336515436170361
Regulation of Transcription by RNA Polymerase II (GO:0006357)	DKK1	0.8331834549403314	0.102802684050163

GO Cellular Component Enrichment table:

Term	Genes	Adjusted P-value	Combined Score
Postsynaptic Density Membrane (GO:0098839)	LRFN5;GRIA4	0.0106364649626829	510.8527085713869

Postsynaptic Specialization Membrane (GO:0099634)	LRFN5;GRIA4	0.01063646496268 29	396.281789679176 5
Postsynaptic Density (GO:0014069)	LRFN5;GRIA4	0.05669885619475 62	94.6813778564908 4
Myosin Filament (GO:0032982)	MYO18B	0.07417640281427 98	421.091823007421 6
AMPA Glutamate Receptor Complex (GO:0032281)	GRIA4	0.07417640281427 98	279.207318677517 66
GABA-ergic Synapse (GO:0098982)	LRFN5	0.07417640281427 98	263.644228828179 8
Ionotropic Glutamate Receptor Complex (GO:0008328)	GRIA4	0.08905907476965 01	158.951830565905 23
Supramolecular Fiber (GO:0099512)	MYO18B	0.08905907476965 01	142.378481464209 13
Extracellular Membrane- Bounded Organelle (GO:0065010)	GRIA4	0.11835132773709 48	77.0124855419272

Extracellular Vesicle (GO:1903561)	GRIA4	0.1183513277370948	72.08123312729496
Glutamatergic Synapse (GO:0098978)	LRFN5	0.1745617480983246	36.6416559464346
Early Endosome Membrane (GO:0031901)	DKK1	0.180948362466265	27.889826667906146
Sperm Flagellum (GO:0036126)	KCNU1	0.180948362466265	27.26710770178677
9+2 Motile Cilium (GO:0097729)	KCNU1	0.1857901318479143	23.47841739926932
Endocytic Vesicle Membrane (GO:0030666)	GRIA4	0.2026110024309492	18.53050889894044
Endocytic Vesicle (GO:0030139)	GRIA4	0.2285374533886652	13.860544185302931
Vesicle (GO:0031982)	GRIA4	0.2436864575559186	11.31845077841186
Dendrite (GO:0030425)	GRIA4	0.2763124237419992	8.314041492443591
Early Endosome (GO:0005769)	DKK1	0.2941611336127798	6.769424474545983

Focal Adhesion (GO:0005925)	TM4SF20	0.30515508098021 79	4.96839356794313 2
Endosome Membrane (GO:0010008)	DKK1	0.30515508098021 79	4.86371360569988 55
Cell-Substrate Junction (GO:0030055)	TM4SF20	0.30515508098021 79	4.80246760932887 45
Cytoplasmic Vesicle Membrane (GO:0030659)	GRIA4	0.30515508098021 79	4.42183544605982 4
Neuron Projection (GO:0043005)	GRIA4	0.37589151872589 1	2.62744319849684 4
Endoplasmic Reticulum Membrane (GO:0005789)	TM4SF20	0.49134524818046 01	1.12541535485212 55
Bounding Membrane of Organelle (GO:0098588)	GRIA4	0.49134524818046 01	1.10196189317483 58

GO Molecular Function Enrichment table :

Term	Genes	Adjusted P-value	Combined Score
AMPA Glutamate Receptor Activity (GO:0004971)	GRIA4	0.03887690641353 26	1993.60946675891 9
Dipeptidase Activity (GO:0016805)	MEP1A	0.03887690641353 26	776.212425317664
Ligand-Gated Monoatomic Ion Channel Activity Involved in Regulation of Presynaptic Membrane Potential (GO:0099507)	GRIA4	0.03887690641353 26	776.212425317664
Disulfide Oxidoreductase Activity (GO:0015036)	PDIA2	0.03887690641353 26	776.212425317664
Calcium-Activated Potassium Channel Activity (GO:0015269)	KCNU1	0.03887690641353 26	776.212425317664
Glutamate-Gated Receptor Activity (GO:0004970)	GRIA4	0.04120580728430 74	421.091823007421 6
Receptor Antagonist Activity (GO:0048019)	DKK1	0.04120580728430 74	389.380440824271 26

Glutamate Receptor Activity (GO:0008066)	GRIA4	0.0412058072843074	389.38044082427126
Low-Density Lipoprotein Particle Receptor Binding (GO:0050750)	DKK1	0.04749817321048	249.55202523941097
Neurotransmitter Receptor Activity Involved in Regulation of Postsynaptic Membrane Potential (GO:0099529)	GRIA4	0.04749817321048	204.47351966705395
Lipoprotein Particle Receptor Binding (GO:0070325)	DKK1	0.04749817321048	195.39226686727991
Ligand-Gated Monoatomic Ion Channel Activity (GO:0015276)	GRIA4	0.04749817321048	179.21238137059572
Potassium Ion Transmembrane Transporter Activity (GO:0015079)	KCNU1	0.04749817321048	153.06869330263612
N-acyltransferase Activity (GO:0016410)	GLYATL2	0.04749817321048	153.06869330263612

Transmitter-Gated Monoatomic Ion Channel Activity Involved in Regulation of Postsynaptic Membrane Potential (GO:1904315)	GRIA4	0.04749817321048	142.37848146420913
Inorganic Cation Transmembrane Transporter Activity (GO:0022890)	GRIA4	0.051642330505899	116.97591748147126
Transmitter-Gated Monoatomic Ion Channel Activity (GO:0022824)	GRIA4	0.0541617180096198	101.22516915069232
Metalloexopeptidase Activity (GO:0008235)	MEP1A	0.0542929680909871	93.44062326514334
Potassium Channel Activity (GO:0005267)	KCNU1	0.0721853054267386	55.80808062680676
Carboxylic Acid Transmembrane Transporter Activity (GO:0046943)	GRIA4	0.0721853054267386	54.81664896212504
Growth Factor Activity (GO:0008083)	DKK1	0.0827595635479745	42.24570226347746

Metalloendopeptidase Activity (GO:0004222)	MEP1A	0.0848020585207704	38.18665714573386
Metallopeptidase Activity (GO:0008237)	MEP1A	0.0965247760985688	28.53621702932384
Monoatomic Cation Channel Activity (GO:0005261)	KCNU1	0.0965247760985688	27.889826667906146
Endopeptidase Activity (GO:0004175)	MEP1A	0.2179419369257462	7.084574495046588
Receptor Ligand Activity (GO:0048018)	DKK1	0.2501209890076788	5.120656068117643

KEGG Pathway Enrichment Analysis via Enrichr:

KEGG 2021 Human pathway enrichment analysis of top 15 ILINCS-derived genes, highlighting pathways such as Wnt signaling, Neurodegeneration, and Insulin secretion. Results are ranked based on combined score and adjusted p-values.

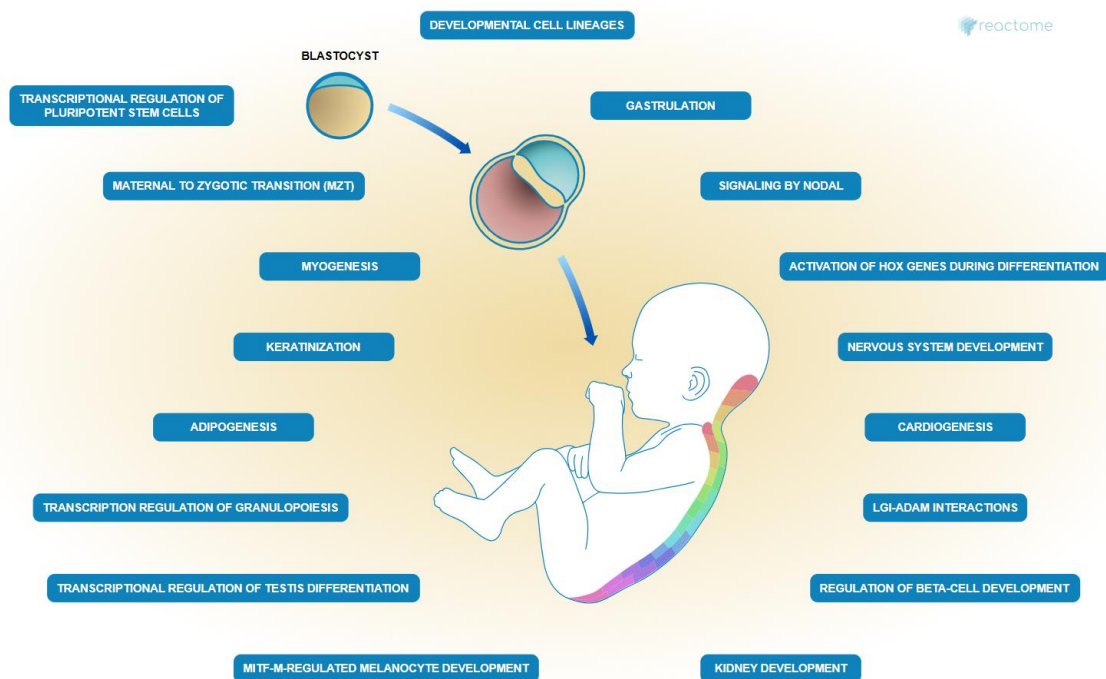
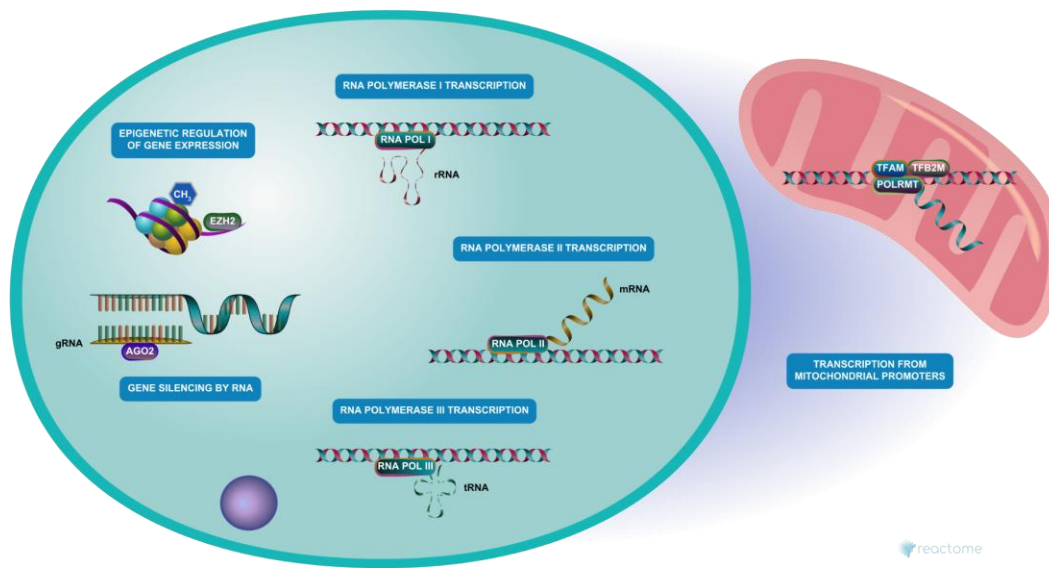
Term	Overlap	P-value Adjusted	P-value	Old P-value	Old Adjusted P-value	Odds Ratio	Combined Score	Genes
Nicotine addiction	1/40	0.02374404479036207	0.1276651238964805	0	0	46.5011655011655	173.93405340895535	GRIA4

Pathways of neurodegeneration	2/475	0.0317372 377989380 84	0.12766 512389 64805	0	0	8.251585 62367864 7	28.4701 537179 35595	DKK1
Amphetamine addiction	1/69	0.0406341 472797855 2	0.12766 512389 64805	0	0	26.63101 60427807 5	85.3030 458253 5365	GRIA4
Insulin secretion	1/86	0.0504102 825788442 1	0.12766 512389 64805	0	0	21.28663 10160427 8	63.5950 895996 09956	KCNU1
Circadian entrainment	1/97	0.0566872 670654609 8	0.12766 512389 64805	0	0	18.83712 12121212 1	54.0664 119251 6822	GRIA4
Protein digestion and absorption	1/103	0.0600950 2	0.12766 512389 64805	0	0	17.72370 76648841 35	49.8360 217038 30525	MEP1A

Glutamatergic synapse	1/114	0.06631328522892102	0.1276651238964805	0	0	15.989541432019308	43.38546242886158	GRIA4
Dopaminergic synapse	1/132	0.07640733078430288	0.1276651238964805	0	0	13.78001387925052	35.437739719643616	GRIA4
Vascular smooth muscle contraction	1/133	0.07696516706764417	0.1276651238964805	0	0	13.674931129476583	35.06802532248599	KCNU1
Retrograde endocannabinoid signaling	1/148	0.08529573	0.1276651238964805	0	0	12.270253555967841	30.204834380712917	GRIA4

Wnt signaling pathway	1/166	0.0952014 210864982 9	0.12766 512389 64805	0	0	10.92176 30853994 5	25.6853 700307 35868	DKK1
cGMP-PKG signaling pathway	1/167	0.0957488 429223603 8	0.12766 512389 64805	0	0	10.85542 16867469 88	25.4671 095006 73597	KCNU1
cAMP signaling pathway	1/216	0.1222035 696832793 5	0.15040 439345 634382	0	0	8.360676 53276955 6	17.5747 024116 17213	GRIA4
Huntington disease	1/306	0.1689537 697970903	0.19309 002262 524602	0	0	5.866766 02086438 15	10.4318 735621 63403	GRIA4

Neuroactive ligand- receptor interaction	1/341	0.1865095 226643763	0.19894 349084 200136	0	0	5.253475 93582887 7	8.82202 019715 2945	GRIA4
Alzheimer disease	1/369	0.2003086 327686292 3	0.20030 863276 862923	0	0	4.846837 94466403 1	7.79321 104358 6478	DKK1

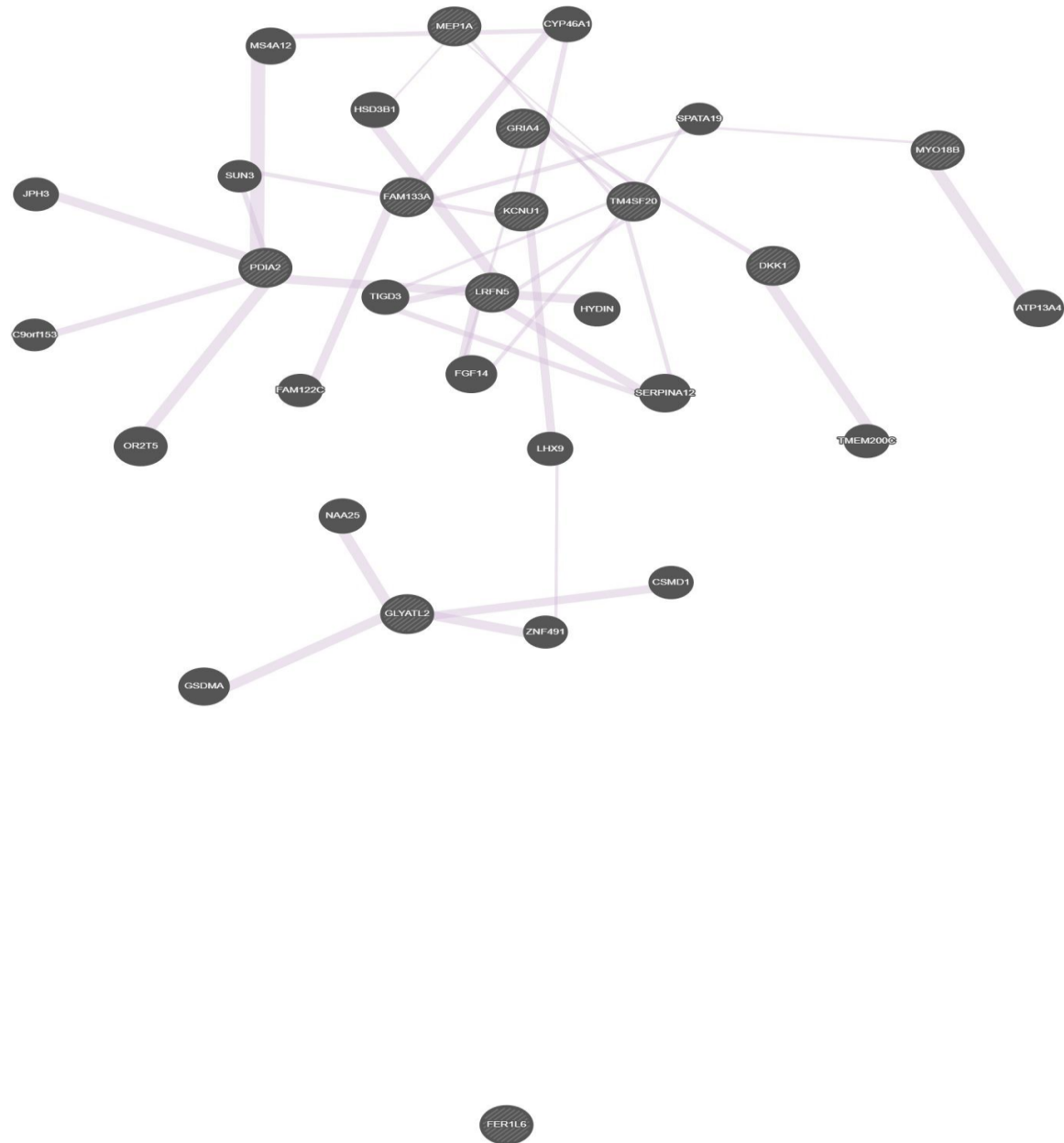


7. Advanced Multi-Ontology Analysis (ToppGene)

ToppGene revealed GO terms related to synaptic assembly, membrane domains, Wnt signaling, and phenotypes such as vertebral fusion. Pathways from Reactome and BioCarta validated links to HSC-associated remodeling.

8. Network Interaction Analysis (GeneMANIA)

GeneMANIA identified co-expressed and functionally predicted networks among key genes like GRIA4, PDIA2, and DKK1. This provides support for their roles in neurosignaling and cancer-specific remodeling.



10. Discussion

The integrative transcriptomic analysis conducted in this study revealed several differentially expressed genes (DEGs) with strong functional associations to biological pathways implicated in hepatobiliary carcinoma (HBC) progression and hepatic stellate cell (HSC) activation.

Among the most significantly upregulated genes, *DKK1* (Dickkopf WNT Signaling Pathway Inhibitor 1) emerged as a central node influencing multiple aspects of Wnt signaling regulation. Enrichment analyses (GO and Reactome) confirmed its strong involvement in non-canonical Wnt signaling, planar cell polarity, and regulation of SMAD protein signal transduction. Given *DKK1*'s role in antagonizing Wnt/ β -catenin signaling, its overexpression suggests a shift in tumor microenvironment dynamics, possibly contributing to immune evasion, fibrosis, and altered differentiation—hallmarks of HSC activation and ECM remodeling. These findings support the hypothesis that *DKK1* overexpression might act as a biomarker of fibrosis-driven carcinogenesis and a therapeutic target for intercepting tumor progression.

Another prominent DEG, *GRIA4* (Glutamate Ionotropic Receptor AMPA Type Subunit 4), was significantly enriched in pathways related to neurotransmission, synaptic plasticity, and glutamate receptor activity. Its expression was linked to the Glutamatergic synapse and dopaminergic signaling, suggesting potential neuroimmune cross-talk within the tumor niche. The presence of *GRIA4* in postsynaptic density membranes and endocytic vesicles further highlights its role in vesicular trafficking and cellular signaling, potentially modulating immune infiltration or tumor cell communication. These findings suggest that altered neurotransmitter signaling may contribute to microenvironmental remodeling in HBC.

KCNU1 (Potassium Calcium-Activated Channel Subfamily U Member 1), involved in potassium ion transmembrane transport, was enriched in insulin secretion and cAMP/cGMP signaling pathways. These pathways are known to influence cellular metabolism and apoptosis regulation. The dysregulation of *KCNU1* might reflect compensatory ionic shifts in tumor cells or HSCs under fibrotic and inflammatory stress.

Furthermore, *PDIA2* and *MEP1A*, involved in endoplasmic reticulum (ER) stress responses and peptide processing, respectively, were linked to protein retention in the ER lumen and dipeptidase activity. Their upregulation indicates a heightened demand for protein folding and degradation processes in the fibrotic tumor microenvironment. This aligns with the known secretory and ECM-producing phenotype of activated HSCs.

Multi-ontology enrichment from ToppGene, along with PPI networks derived from GeneMANIA, supported these findings by revealing tightly connected modules involved in Wnt signaling, vesicle transport, synaptic assembly, and ER stress—biological themes central to both tumor aggressiveness and HSC-mediated fibrosis.

Collectively, these results suggest a multi-dimensional alteration of signaling and structural pathways that facilitate hepatobiliary tumor progression. We hypothesize that:

- Upregulated DKK1 disrupts canonical Wnt signaling, contributing to dedifferentiation and fibrosis.
- GRIA4-mediated neurotransmission alterations may support immunosuppressive microenvironments.
- KCNU1 and MEP1A dysregulation reflect shifts in ionic balance and proteostasis, essential for cellular survival under tumor-induced stress.

These gene signatures not only distinguish HCC from CCA but also reflect the pathological state of the surrounding microenvironment, offering candidates for biomarker development and targeted therapy. Future wet-lab validation and single-cell resolution studies can further dissect the stromal versus epithelial contributions of these DEGs.

11. Conclusion

This study uncovers key gene expression signatures—*DKK1*, *GRIA4*, *KCNU1*, *PDIA2*, and *MEP1A*—that are closely linked to hepatic stellate cell (HSC) activation and tumor microenvironment remodeling in hepatobiliary carcinomas (HBCs). These genes participate in pathways related to fibrosis, immune regulation, Wnt signaling, neurotransmission, and ER stress, highlighting their roles in disease progression and therapeutic resistance.

Our findings emphasize the interplay between tumor cells and stromal components, offering new avenues for biomarker development and stroma-targeted therapies. By using publicly available transcriptomic data and reproducible bioinformatics pipelines, this study supports open-science approaches in advancing precision oncology.

Future research should validate these targets in patient tissues and explore interventions—such as small molecules or CRISPR—to modulate HSC-driven tumor dynamics. Integration of spatial and single-cell transcriptomics could further reveal microenvironmental heterogeneity, improving diagnosis and treatment of HBCs.

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